Yuyang ZHAO

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EDUCATION

School of Qiushi Honors, Tianjin University, China *Sept 2016 - Jun 2020*

- **Bachelor of Engineering** in Electronic Information Engineering
- **Overall GPA:** 3.83/4.0.
- **Core Courses:** Advanced Mathematics (98 for the first term, 93 for the second term), Linear Algebra and Application (95), Probability Theory and Mathematic Statistics (98), Computer Networking (93), Digital Image Processing (95).
- **Honors&Prize:** Outstanding undergraduate student of Tianjin University;
  Third Prize of Asia and Pacific Mathematical Contest in Modeling.

PUBLICATIONS

- **Yuyang Zhao**, Zhun Zhong, Fengxiang Yang, Zhiming Luo, Yaojin Lin, Shaozi Li, Nicu Sebe
  Learning to Generalize Unseen Domains via Memory-based Multi-Source Meta-Learning for Person Re-Identification
  Accepted by **IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2021)**
  [PAPER]
- Xiaolin Song, **Yuyang Zhao**, Jingyu Yang, Cuiling Lan, Wenjun Zeng
  FPCR-Net: Feature Pyramidal Correlation and Residual Reconstruction for Semi-supervised Optical Flow Estimation
  Submitted to **Neurocomputing**
  [PAPER]
- Xiaolin Song, **Yuyang Zhao**, Jingyu Yang
  STC-Flow: Spatio-temporal Context-aware Optical Flow Estimation
  Submitted to **Signal Processing: Image Communication**
  [PAPER]
- Jinsong Zhang, **Yuyang Zhao**, Kun Li, YeBin Liu, Jingyu Yang, Qionghai Dai
  Attention-guided GANs for human pose transfer
  Published on **SPIE 11187, Optoelectronic Imaging and Multimedia Technology VI**
  [PAPER]

RESEARCH EXPERIENCES

Intelligent Multimedia Technology Lab, Xiamen University

Research Assistant *Jul 2020 - Present*

Advisor: Shaozi Li, Professor at Cognitive Science Department

Co-Advisor: Zhiming Luo, Postdoc Researcher at Cognitive Science Department

Co-Advisor: Zhun Zhong, Postdoc Researcher at the University of Trento

Project: Person Re-Identification

- Proposed a Multi-Source Meta-Learning framework for domain generalization in person re-identification.
- Proposed a non-parametric memory-based identification loss function to prevent unstable meta-learning optimization.
- Presented a novel meta batch normalization layer (MetaBN) to generate more diverse features for meta-learning and to improve the generalization ability.
- Outperformed other state-of-the-art methods on four large-scale datasets.
- The paper summarizing this work was accepted by CVPR2021.

Lab of Intelligent Imaging and Reconstruction, Tianjin University

Research Assistant *Sept 2019 – Jun 2020*

Advisor: Jingyu Yang, Professor at School of Electrical and Information Engineering

Project I: Optical Flow Estimation (STC-Flow)

- Utilized pyramidal processing and correlation to predict the optical flow.
- Applied three blocks to gather long-range contextual information to improve feature representation in the feature extraction stage, the optical flow estimation stage, and the reconstruction stage for optical flow estimation.
- Achieved the lowest average endpoint error on MPI Sintel dataset.
- Submitted the paper summarizing this work to Signal Processing: Image Communication.

Project II: Optical Flow Estimation (FPCR-Net)

- Proposed a correlation-warping-normalization (CWN) module to exploit the correlation dependency for optical flow estimation.
- Proposed a residual reconstruction module to reconstruct the sub-band high-frequency information and improve the details and textures at each stage.
- Proposed a novel semi-supervised loss function for optical flow estimation.
- Achieved the lowest average endpoint error on MPI Sintel dataset final pass and KITTI 2012 dataset.
- Submitted the paper summarizing this work to Neurocomputing

The Lab for Graphics, Vision and Media Computing, Tianjin University

Research Assistant *Jul 2019 – Sept 2019*

Advisor: Kun Li, Associate Professor at College of Intelligence and Computing

Project: Human Pose Transfer

- Applied conditional GANs to transfer a person in one pose to another pose.
- Applied a new block —Pose-Guided Attention Block (PGAB) to fuse poses and images.
- Achieved the highest IS on the Market-1501 dataset and the highest SSIM on the DeepFashion dataset among all the methods at the time of writing.
- Published the paper summarizing this work on SPIE 11187, Optoelectronic Imaging and Multimedia Technology VI.

Research Assistant *May 2019 – Jul 2019*

Advisor: Bo Wu, Postdoc Researcher at Columbia University

Project: Reinforcement Learning in Playing Atari Games

- Applied reinforcement learning in the Atari games.
- Applied different algorithms of reinforcement learning, including Deep Q Network, Double Deep Q Network, Dueling Deep Q Network, and Asynchronous Advantage Actor-critic.
- Applied different models and loss functions for these algorithms.

Research Assistant *Apr 2019 – May 2019*

Advisor: Qingqiu Huang, Ph.D. at The Chinese University of Hong Kong

Project: Shot Boundary Detection

- Applied a traditional method to locate a shot.
- Calculated the difference of a sequence of frames to find the possible boundary of a shot.
- Optimized the indices of possible frames by comparing them with previous and subsequent several frames.

LEADERSHIP EXPERIENCES

Student Association for Science and Technology, Tianjin University | Minister *Sept 2017 – Jul 2018*

- Designed publicity materials for almost all the club activities in the Student Association for Science and Technology, including but not limited to the posters of the Student Science Prize, and the brochures of the 'Intelligent Space Station' Seminar.

SKILLS

Programming Languages & Software: Experienced in Matlab, Python (including Pytorch), LaTeX (Revised the LaTex Template of the 2020 Undergraduate Thesis of Tianjin University, [CODE])

Language: TOEFL: Total 104; GRE: Total 322 + 3.5